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Processing of the precursor to corticotropin and β-lipotropin in humans

(immunoprecipitations/peptide analysis/peptide hormones/Nelson syndrome/ectopic corticotropin)

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ABSTRACT **The biosynthesis of human corticotropin (ACTH) was studied in organ culture of pituitary adenomas and by translating mRNA from an ectopic ACTH-producing tumor in a cell-free system. Peptides similar to human ACTH, β-lipotropin, and the amino-terminal glycopeptide are cleaved from a common precursor with an apparent molecular weight of 35,000 on sodium dodecyl sulfate/polyacrylamide gels. The precursors synthesized in pituitary and ectopic ACTH-producing tissues are indistinguishable. The cleavage sites of the peptide chain appear to be similar to those previously deduced for murine and bovine ACTH. Immunoprecipitation studies suggest that the primary structure of the precursor peptide is similar in all three species. However, glycosylation is different in the human and murine precursors: the precursor to human ACTH appears to be glycosylated only in the amino-terminal fragment, not in the ACTH or β-lipotropin sequences. Studies with an autopsied normal human pituitary suggest that neither normal nor adenomatous pituitary tissue glycosylates the ACTH sequence.**

In several animal species, corticotropin (ACTH), β-lipotropin (β-LPH), endorphin, and melanotropin (MSH, melanocyte-stimulating hormone) arise from a common precursor protein (pro-ACTH/LPH) (1–9). In the cultured mouse pituitary tumor line AtT20, a precursor protein of apparent M_r 30,000 is cleaved at two places to yield ACTH, β-LPH, and a large amino-terminal fragment lacking known biological function (1–4). In certain tissues, further cleavage of ACTH yields α-MSH and corticotropin-like intermediate lobe peptide; the β-LPH sequence may be cleaved to yield various peptides, including γ-LPH, β-MSH, and β-endorphin. The precursor protein and its major cleavage products other than β-LPH may be glycosylated in various ways prior to cleavage (10, 11). In the intermediate lobe of bovine pituitary, ACTH, β-LPH, and an amino-terminal region of ≈105 amino acids arise from a common precursor protein whose sequence was deduced from the mRNA nucleotide sequence (8, 9).

Less information exists about the genesis and processing of human ACTH, but it suggests a similar scheme. Plasma concentrations of immunoassayable pituitary ACTH and β-LPH increase and decrease concomitantly in normal and pathologic states (12–14). Immunoassayable β-LPH and endorphin are found along with ACTH in pancreatic (15) and pulmonary (16) tumors causing the ectopic ACTH syndrome. High-molecular-weight placental proteins prepared by immunoadsorption to antiserum to ACTH-(1–24) crossreact with antibodies to β-LPH and endorphin (17). However, it is not known if the processing of pro-ACTH/LPH (i.e., its glycosylation and proteolytic cleavage) is similar in mice and humans. Furthermore, it is not known if pituitary and ectopic ACTH arise from the same precursor synthesized from the same gene.

In order to elucidate the pathways of processing of human pro-ACTH/LPH, we studied an ectopic ACTH-producing tumor by immunoprecipitation and tryptic mapping of pro-ACTH/LPH coded by tumor mRNA in a cell-free protein-synthesizing system. We also studied the incorporation of [^{35}S]methionine and [^{3}H]glucosamine in organ culture of human pituitary tumors and measured ACTH in peptides extracted from a human pituitary post mortem. Our data indicate that human ACTH arises from a precursor of apparent M_r 35,000 which appears to be the same in pituitary and ectopic ACTH-producing tumors. This precursor is cleaved by a proteolytic process indistinguishable from that seen in AtT20 cells. However, unlike in AtT20 cells, only the amino-terminal fragment of the human precursor is significantly glycosylated.

MATERIALS AND METHODS

RNA Preparation and Translation. Metastatic tissue from an ectopic ACTH-producing carcinoma of the pancreas from a 10-year-old boy was frozen in liquid nitrogen at surgery and stored at −70°C until used. Frozen tissue was pulverized and dissolved in 5 M guanidine thiocyanate/50 mM Tris·HCl, pH 7.5/10 mM EDTA/5% (wt/vol) 2-mercaptoethanol/1% sodium lauroyl sarcosinate containing 0.45 g of CsCl per ml for preparation of RNA by centrifugation through a cushion of 5.7 M CsCl (18). Polyadenylylated RNA [poly(A)-RNA] was selected from the total RNA by chromatography on oligo(dT)-cellulose (19). The poly(A)-RNA was translated in a cell-free rabbit reticulocyte system treated with micrococcal nuclease to destroy endogenous mRNA (20).

Immunoprecipitations. Immunoprecipitations from the cell-free translation products were performed by modifications of the method of Kessler (21) as described (1, 22). All antibodies were raised in rabbits by E. Herbert, except for the anti-LPH, provided by C. H. Li. Antibody against ACTH-(1–24) was purified by affinity chromatography against α-ACTH-(1–24) bound to Sepharose 4B (3), and antibody against endorphin was raised against purified conjugated endorphin; neither antibody contained material that crossreacted with other regions of the ACTH/LPH precursor (unpublished observations). Anti-LPH crossreacted with β-endorphin (23). Two antisera raised against

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Abbreviations: ACTH, corticotropin (adrenocorticotropic hormone); LPH, lipotropin (lipotropic hormone); hpro-ACTH/LPH, precursor to human ACTH and LPH with apparent M_r 35,000; MSH, melanotropin (melanocyte-stimulating hormone).

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proteins secreted from AtT20 cells reacted with the amino-terminal region of the ACTH/LPH precursor. B-FT is antibody from which anti-ACTH-(1–24) was removed by affinity chromatography; it also reacted with β-LPH; antibody Bgt reacted with ACTH-(1–39), β-LPH, and endorphin, as well as with the amino-terminal fragment.

***In Vitro* Tissue Labeling.** A 1-mg sample of pituitary adenoma from a 12-year-old girl with Nelson syndrome [ACTH-producing tumor of the pituitary after bilateral adrenalectomy in a patient with Cushing disease (24)] was obtained at surgery and incubated for 2.5 hr at 37°C in 100 μl of Dulbecco's minimal essential medium deficient in methionine and cysteine and supplemented with 80 μCi of [^{35}S]methionine (New England Nuclear; 1075 Ci/mmol; 1 Ci = 3.7×10^{10} becquerels). For carbohydrate labeling, a sample of tumor from a 19-year-old man with Cushing disease was incubated for 6 hr in the same medium deficient in glucose and supplemented with 1.0 mCi of [^{3}H]glucosamine (New England Nuclear; 20 Ci/mmol). After incubation, the tissues were washed in phosphate-buffered saline free of calcium and magnesium and lysed in 50 μl of isoelectric focusing lysis buffer (25).

Gel Electrophoresis and Tryptic Peptide Analysis. One-dimensional electrophoresis was performed in $NaDodSO_4$ gels containing 12.5% (wt/vol) polyacrylamide as described by Laemmli (26). For two-dimensional electrophoresis of total cell proteins, the first dimension consisted of nonequilibrium pH gradient gel electrophoresis with pH 3.5–10 Ampholines for 1600 V·hr, and the second dimension consisted of $NaDodSO_4$ gel electrophoresis in 12.5% polyacrylamide (27).

Portions of dried gels were rehydrated in 0.1 M NH_4HCO_3 after autoradiography, and immunoprecipitates or peptides labeled *in vivo* were eluted by digestion with tosylphenylalanyl chloromethyl ketone-treated trypsin (25 μg/ml) for 18 hr at 37°C (28). Tryptic peptides were analyzed by paper electrophoresis at pH 6.5 as described (1, 4).

Normal Pituitary ACTH Profile. A pituitary from a patient who died from bronchopneumonia was obtained 3 hr after death. After the tissue was extracted with 5 M acetic acid, the acid-soluble proteins were electrophoresed on a 12% acrylamide/$NaDodSO_4$ tube gel, protein was eluted from gel slices, and ACTH was measured by radioimmunoassay with antibody against ACTH-(1–24) (10).

RESULTS

Cell-Free Synthesis of pro-ACTH/LPH. Metastatic tissue (15 g) from the ectopic ACTH-producing tumor contained 44.3 mg of extractable RNA, of which 2.3 mg was polyadenylylated. This poly(A)-RNA stimulated efficient incorporation of [^{35}S]-methionine into acid-precipitable protein in the rabbit reticulocyte cell-free translation system. ACTH precursor was immunoprecipitated from the translation products and displayed on $NaDodSO_4$/polyacrylamide gels. An ACTH precursor with an apparent M_r ≈35,000 was precipitated by antibodies to ACTH-(1–24) (Fig. 1, lane D), LPH (Fig. 1, lane B), and endorphin (Fig. 1, lane C) but not by antibody to growth hormone or by control serum lacking ACTH antibodies (not shown). The band at M_r 35,000 is faint because mRNA coding for pro-ACTH/LPH constituted only about 0.3% of the total mRNA. Immunoprecipitation of this protein was specifically blocked when ACTH-(1–39) was added in 1000-fold excess to the reaction mixtures with anti-ACTH-(1–24) (Fig. 1, lane E). Similar inhibition of specific immunoprecipitation was obtained with excess β-LPH or endorphin when the immunoprecipitations were performed with anti-LPH or anti-endorphin, respectively (data not shown). The human pro-ACTH/LPH molecule of M_r 35,000 (hereafter referred to as hpro-ACTH/LPH), however,

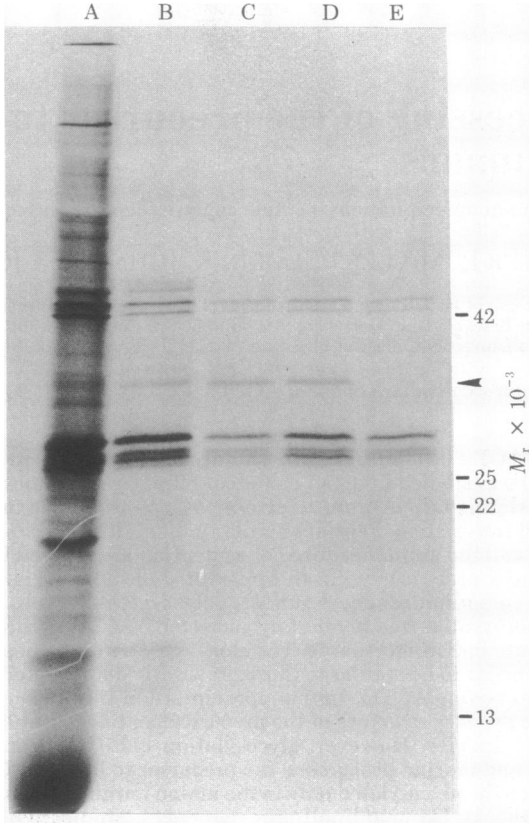


FIG. 1 $NaDodSO_4$/polyacrylamide gel electrophoresis of immunoprecipitated ACTH. mRNA from an ectopic ACTH-producing tumor was translated *in vitro* and the ACTH-related peptides were immunoprecipitated with *Staphylococcus aureus* Cowan strain I. Immunoprecipitates or total cell-free translation products were displayed on 12.5% gels and the [^{35}S]methionine-labeled proteins were visualized by autoradiography. Lane A, total translation products; lane B, proteins immunoprecipitated by anti-LPH; lane C, proteins immunoprecipitated by anti-endorphin; lane D, proteins immunoprecipitated by anti-ACTH-(1–24); lane E, as lane D with exogenously added ACTH-(1–39). Approximate M_r was determined from the migration of nonradioactive markers in another lane of the same gel (ovalbumin, 42,000; chymotrypsin, 25,000; trypsin, 22,000; and ribonuclease, 13,000).

was not precipitated by antibodies directed toward the amino-terminal region of the murine ACTH precursor. Thus, the human and murine amino-terminal regions may contain different antigenic determinants. The two families of bands with M_r ≈ 28,000–30,000 and 44,000–45,000 seen in all lanes were due to nonspecific immunoprecipitation and were immunoprecipitated by antibodies to growth hormone and by control serum lacking ACTH antibodies. Furthermore, their precipitation was not blocked by added excess ACTH. Thus, antibodies specific to ACTH-(1–24), LPH, and endorphin all precipitate the same M_r 35,000 protein from the cell-free translation products coded by mRNA from an ectopic ACTH-producing tumor.

Cellular Synthesis of pro-ACTH/LPH. Proteins from the ACTH-producing pituitary adenomas labeled *in vitro* with either [^{35}S]methionine or [^{3}H]glucosamine were displayed on two-dimensional gels (Fig. 2). In contrast to the ectopic ACTH-producing tumor, hpro-ACTH/LPH was an abundant protein in the pituitary tumors. The large spot (A) dominating the center of Fig. 2 *Upper* was identified as hpro-ACTH/LPH by several techniques. First, its apparent M_r was 35,000, as determined by comparison to internal standards of known M_r. Second, its location with respect to other protein spots was

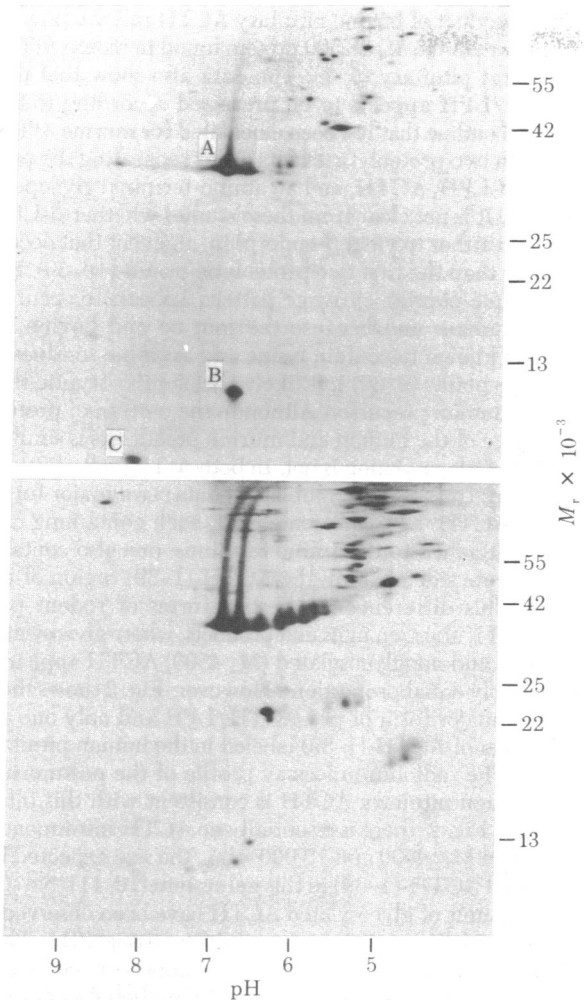


FIG. 2. Autoradiograph of two-dimensional gel electrophoresis of proteins synthesized by ACTH-producing pituitary tumors labeled with [^{35}S]methionine or [^{3}H]glucosamine. The tissue was incubated in medium containing [^{35}S]methionine (*Upper*) or [^{3}H]glucosamine (*Lower*), lysed, layered onto a tube gel, and subjected to two-dimensional gel electrophoresis. M_r and pH were measured from a parallel gel handled in the same fashion displaying nonradioactive proteins of known size. A, hpro-ACTH/LPH; B, β-LPH; C, ACTH-(1–39).

analogous to the M_r 29,000 and 32,000 glycosylated murine ACTH precursor from labeled AtT20 cells when run on similar gels (29). Third, immunoprecipitated hpro-ACTH/LPH run on a two-dimensional gel had identical mobility. Fourth, the electrophoretic pattern of tryptic peptides from the protein in this spot was similar to that for the murine pro-ACTH/LPH (see below). The prominent spot with M_r ≈11,700 and migrating at a pH similar to that of the hpro-ACTH/LPH (B) appears to be β-LPH, whereas the ACTH-(1–39) migrated toward the more basic end of the gel with a M_r 4500 (C). The identity of these spots as β-LPH and human ACTH of M_r 4500 was determined by the same four techniques used for human ACTH of M_r 35,000. Thus, both the isoelectric points and the apparent M_r of the human ACTH precursor and its end products are similar to those in the mouse. The human pattern, however, showed less of the heterogeneity about the 35,000 M_r precursor seen in the mouse AtT20 precursors (29). What heterogeneity exists was not seen in the 4500 M_r ACTH and the β-LPH molecules. Close inspection of the gel, however, showed that the pattern of satellite spots seen about the 35,000 M_r human ACTH precursor was also present at a pH of about 6.0 at a M_r of about 24,000, probably representing ACTH-(1–39) linked to the amino-terminal fragment after the β-LPH sequence was cleaved off. A similar array of spots was also seen at a pH of about 5.5 at a M_r of about 18,000, probably representing the glycosylated amino-terminal fragment itself. This heterogeneity of M_r and pI is due to heterogeneity in the glycosylation of the amino-terminal fragment.

These assignments are supported by the [^{3}H]glucosamine incorporation experiment done in tumor tissue from a patient with Cushing disease (Fig. 2 *Lower*). The longer labeling period of this experiment (6 hr vs. 2.5 hr) may account for the greater heterogeneity about the M_r 35,000 ACTH spot, although this is a somewhat different tumor. The chain of discrete spots of increasing M_r and decreasing pI reflects further addition of glucosamine and sialic acid residues. After cleavage of the 92-amino-acid β-LPH moiety, ACTH-(1–39) remained connected to the amino-terminal fragment, as seen at M_r 22,000–24,000 at pH ≈ 6. The nonglycosylated ACTH moiety was then cleaved, leaving the amino-terminal fragment (M_r 16,000–18,000, pH ≈ 5). The group of spots at M_r 22,000–24,000 and at M_r 16,000–18,000 are not so dark as the group at M_r 35,000 because the 6-hr incubation did not achieve steady-state labeling (24 hr in AtT20 cells).

Tryptic Peptide Analysis. Proteins coded by the poly(A)-RNA from the ectopic ACTH-producing tumor and translated by reticulocyte cell lysate were immunoprecipitated by anti-ACTH-(1–24) and displayed by $NaDodSO_4$/acrylamide gel electrophoresis as in Fig. 1. The region of the gel corresponding to the hpro-ACTH/LPH band was cut out, the proteins were eluted by digestion with trypsin, and the tryptic peptides were displayed by paper electrophoresis. The distribution of [^{35}S]-methionine in the tryptic peptides of the hpro-ACTH/LPH (Fig. 3) was essentially the same as in tryptic peptides of the mouse ACTH precursor of M_r 28,500 (figure 3*C* in ref. 1). Regions of the two-dimensional gels of proteins synthesized by the human pituitary adenoma tissue were processed in the same way. The tryptic map of the pituitary hpro-ACTH/LPH (Fig. 3) was indistinguishable from the map of the ectopic hpro-ACTH/LPH (Fig. 4*A*). Both hpro-ACTH/LPH maps (Figs. 3 and 4*A*) and the β-LPH map (Fig. 4*B*) contain a prominent peak a short distance from the origin at a relative mobility of 0.05 with respect to ε-dinitrophenyllysine (neutral pH). The

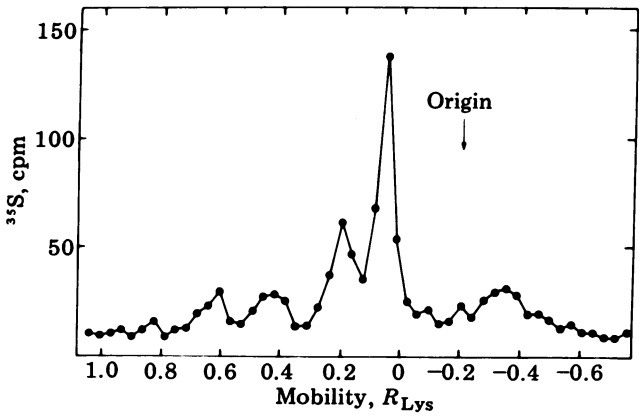


FIG. 3. Paper electrophoresis of hpro-ACTH/LPH tryptic peptides from immunoprecipitates. The hpro-ACTH/LPH synthesized *in vitro* from ectopic tumor mRNA was immunoprecipitated with anti-ACTH-(1–24) and displayed on gels as in Fig. 1. Gel fragments containing the hpro-ACTH/LPH were digested with trypsin and the resulting peptides were electrophoresed on paper with 10% (wt/vol) pyridine at pH 6.5. Radioactivity in 8-mm slices (minus background) is plotted relative to the mobility of lysine. Lysine, ε-dinitrophenyllysine, and picric acid were included as internal markers.

map of human ACTH of M_r 4500 (Fig. 4C) contains a single peak at R_{Lys} 0.19, in the same location as major peaks in the mouse ACTH of M_r 28,500 and hpro-ACTH/LPH maps. The human pro-ACTH/LPH map also contains minor peaks at R_{Lys} about 0.61 and 0.42 and possibly at R_{Lys} 0.32, which may be related to the minor peaks in the map of mouse ACTH of M_r 28,500 at R_{Lys} 0.4 and 0.6. Thus it appears that all the methionine-labeled tryptic peptides found in ACTH and β-LPH, as well as others, are found in the hpro-ACTH/LPH.

Glycosylation of ACTH in Normal Human Pituitary. Pituitary proteins isolated by acid extraction were separated by $NaDodSO_4$/polyacrylamide electrophoresis and examined for ACTH activity by radioimmunoassay. Recovery from the gel was 87% of the applied immunoactivity. Of the recovered material, 2% was in the region of M_r 18,000–50,000, 4% was in the region of M_r 6000–18,000 [where glycosylated α-ACTH-(1–39) would appear], and 94% was in a peak that comigrated with α-ACTH-(1–39). Because the glycosylation site in murine ACTH is believed to be in the carboxy-terminal portion of the α-ACTH-(1–39) sequence (10, 11), we doubt that glycosylation, if it did occur, would interfere with ACTH immunoactivity as measured with anti-α-ACTH-(1–24).

DISCUSSION

Our studies show that the primary translation product of human pro-ACTH/LPH mRNA is a protein of M_r 35,000 in both pituitary and ectopic tumor tissue. This is similar in apparent size both to one form of camel ACTH (5) and to the primary translation product of bovine pituitary ACTH mRNA (Fig. 1) (8) but larger than the M_r 28,500 protein found in mouse AtT20 cells (1) or rat pituitary (6, 7). The data also show that this hpro-ACTH/LPH appears to be processed according to the same general outline that has been elucidated for murine AtT20 cells wherein two proteolytic events occur separating the precursor into β-LPH, ACTH, and an amino-terminal glycopeptide (10, 30). It is not clear from these studies whether β-LPH is processed further to yield β-endorphin, an event that occurs more slowly than the first two proteolytic events (10, 30). Because the same peptide cleavage pattern appears to occur in mice and humans and because the murine and bovine sequences are known to contain lysine and arginine residues at the cleavage points (9, 30) it is likely that similar tryptic sites exist in the human precursor. Although the pattern of proteolytic cleavage of the human and murine precursors is similar, the pattern of glycosylation is not. In both AtT20 cells (29) and rat pituitaries (ref. 6 and unpublished data) two major forms of the pro-ACTH/LPH molecule exist, each containing carbohydrate attached to the amino terminus; one also contains a carbohydrate side chain in the ACTH-(1–39) region of the precursor This difference in the two forms of rodent pro-ACTH/LPH is also seen in its end products, where glycosylated (M_r 13,000) and nonglycosylated (M_r 4500) ACTH appear in approximately equal proportions. However, Fig. 2 shows there is only one major form of pro-ACTH/LPH and only one detectable form of ACTH-(1–39) labeled in the human pituitary adenoma. The radioimmunoassay profile of the postmortem normal human pituitary ACTH is consistent with this interpretation of Fig. 2; there is essentially no ACTH immunoactivity between M_rs 9000 and 15,000—i.e., the size expected for glycosylated ACTH-(1–39) in this gel system (10, 11). No substantial amounts of glycosylated ACTH have been observed in autopsied and biopsied human pituitary tissue (31). Thus, glycosylation of human pro-ACTH/LPH appears to occur only in the amino-terminal fragment, creating a heterogeneity of M_r and pI in the precursor. This glycosylation explains the array of "satellite spots" seen on the autoradiograph on the acidic side of the hpro-ACTH/LPH spot; this array is also seen at $M_r \approx$ 22,000 (pH 6.0), the putative ACTH/amino-terminal intermediate, and at $M_r \approx$ 18,000 (pH 5.0), the putative amino-terminal fragment.

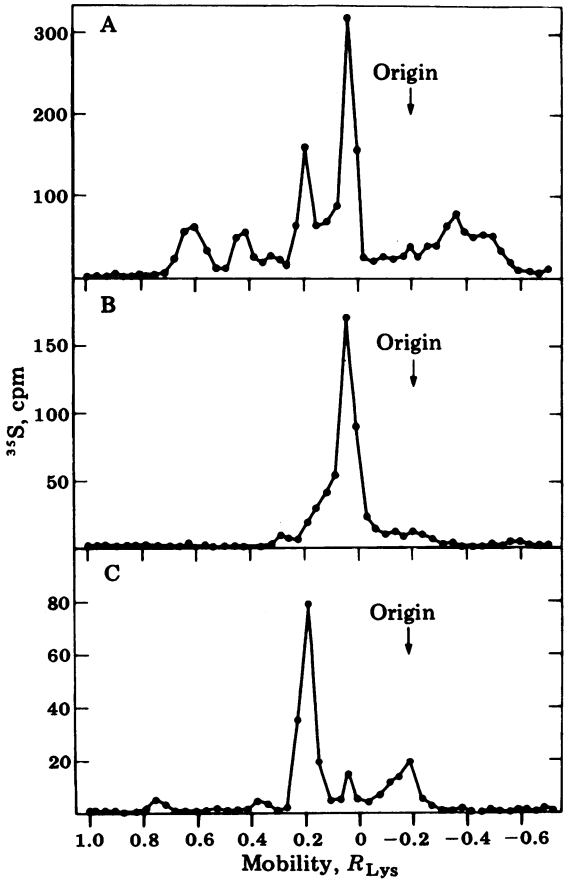


FIG. 4. Paper electrophoresis of tryptic peptides from proteins synthesized by the pituitary adenoma. The spots labeled A, B, and C in Fig. 2 were cut from the gel and the proteins were digested with TPCK-trypsin. Tryptic peptides were displayed as in Fig. 3 (*A*) hpro-ACTH/LPH; (*B*) β-LPH; (*C*) ACTH-(1–39).

The human and murine pro-ACTH/LPH molecules are similar antigenically as well as being similar in their general processing. Antisera directed against murine ACTH-(1–24), β-LPH, and endorphin immunoprecipitate the human peptides, suggesting considerable amino acid homology between the two species in the carboxy-terminal half of the precursor molecule. However, two different antisera, B-FT and Bgt, directed against the amino-terminal region of murine pro-ACTH/LPH of M_r 29,000–32,000 failed to precipitate the hpro-ACTH/LPH, suggesting that there may be different antigenic determinants in the amino-terminal regions of the murine and human precursors. Although unlikely, it is possible that all the antibody reacting with the amino-terminal region in these preparations is directed against a portion of the amino-terminal glycopeptide such as the "γ-MSH" region (9) that is immunologically similar in β-LPH and α-ACTH and that the excess β-LPH or ACTH (which contain MSH regions) in both assays may have competed for the antibody, thus preventing immunoprecipitation of the hpro-ACTH/LPH.

When labeled with [^{35}S]methionine, the tryptic peptides present in the hpro-ACTH/LPH produced by cell-free translation of pancreatic tumor mRNA (Fig. 3) are indistinguishable from the peptides present in the hpro-ACTH/LPH synthesized by the pituitary adenoma (Fig. 4*A*), suggesting that pituitary

and ectopic ACTH arise from very similar precursors. Tryptic mapping of the murine pro-ACTH/LPH by the same procedures yields a similar pattern; however, one less tryptic peptide is observed (1, 2, 10). Human β-LPH contains two tryptic peptides labeled with methionine, β-LPH-(47–51) and β-LPH-(61–69), which are unresolved as a single peak with R_{Lys} 0.05 (Figs. 3 and 4 *A* and *B*). By contrast, mouse and rat β-LPH contain only one methionine-labeled tryptic peptide, β-LPH-(61–69) (2, 32). Like all other mammalian ACTHs of known sequence, human ACTH contains only one methionine-labeled tryptic peptide, α-ACTH-(1–8), represented by a peak with R_{Lys} 0.19 (Figs. 3 and 4 *A* and *C*). In Fig. 4*A*, the peak of β-LPH peptides at R_{Lys} 0.05 contains 720 cpm and the ACTH peak at R_{Lys} 0.19 contains 340 cpm, a ratio of 2.1, in agreement with the assignment of two and one methionine residues to β-LPH and ACTH, respectively. The two peaks at R_{Lys} 0.4 and 0.6 are similar to those observed in the AtT20 system, where they were shown to derive from the amino-terminal region of pro-ACTH/LPH (2). The absence of this material in ACTH (Fig. 4*C*) and β-LPH (Fig. 4*B*) also suggests that this activity derives from the amino terminus. Furthermore, this region contains 350 cpm, suggesting that it represents variable cleavage of a peptide containing a single methionine residue; the amino-terminal fragment of bovine pro-ACTH/LPH, the only species for which sequence data exist, contains a single methionine (9). The origin of the material migrating in the acidic region ($R_{Lys} < 0$) in Figs. 3 and 4*A* is unknown. Thus, analysis of the electrophoretic pattern of tryptic fragments from hpro-ACTH/LPH shows that this material contains the ACTH and β-LPH sequences and further suggests that the human and murine precursors are similar. These data suggest that proteolytic processing of the peptide backbone of the human ACTH precursor in these tumors is similar to that observed in the mouse and rat. Preferred tryptic cleavages appear to occur at certain loci, yielding the active fragments of the precursor molecule. Thus, the pattern of ACTH processing elucidated from mouse AtT20 cells may be a general mammalian feature.

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